

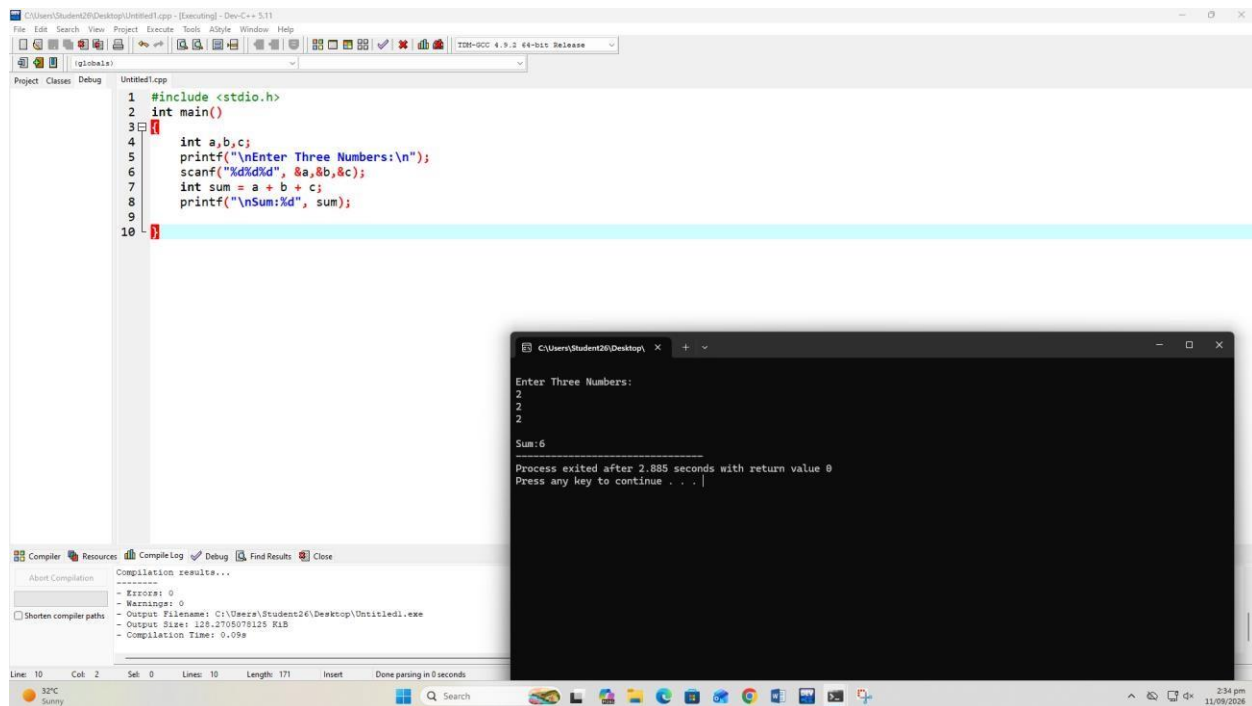
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Section: BSE-1B

IN LAB:

TASK:1



```
1 #include <stdio.h>
2 int main()
3 {
4     int a,b,c;
5     printf("\nEnter Three Numbers:\n");
6     scanf("%d%d%d", &a,&b,&c);
7     int sum = a + b + c;
8     printf("\nSum:%d", sum);
9 }
10
```

Enter Three Numbers:
2
2
2
Sum:6

Process exited after 2.885 seconds with return value 0
Press any key to continue . . .

Compilation results...

- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\Student26\Desktop\Untitled1.exe
- Output Size: 128,270,507,812,5 KIB
- Compilation Time: 0.09s

TASK:2

The screenshot shows a C++ IDE with a file named 'Untitled1.cpp'. The code is as follows:

```
1 #include <stdio.h>
2 int main()
3 {
4     int a,b,c;
5     printf("\nEnter 1st Numbers:");
6     scanf("%d", &a);
7     printf("Enter 2nd Numbers:");
8     scanf("%d", &b);
9     printf("Enter 3rd Numbers:");
10    scanf("%d", &c);
11
12    if (a>b && a>c){
13        printf("%d is Greater", a);
14    }
15    else if (b>a && b>c){
16        printf("%d is Greater", b);
17    }
18    else {
19        printf("%d is Greater", c);
20    }
21 }
```

The output window shows the following text:

```
Enter 1st Numbers:12
Enter 2nd Numbers:21
Enter 3rd Numbers:23
23 is Greater

Process exited after 4.238 seconds with return value 0
Press any key to continue . . .
```

The IDE status bar shows 'Line: 8 Col: 21 Set: 0 Lines: 21 Length: 366 Insert Done parsing in 0.016 seconds'.

TASK:3

The screenshot shows a C++ IDE with a file named 'Untitled1.cpp'. The code is as follows:

```
1 #include <stdio.h>
2 int main()
3 {
4     int num;
5     printf("Enter: ");
6     scanf("%d", &num);
7     switch(num){
8     case 1:
9         printf("\nMonday");
10        break;
11     case 2:
12        printf("\nTuesday");
13        break;
14     case 3:
15        printf("\nWednesday");
16        break;
17     case 4:
18        printf("\nThursday");
19        break;
20     case 5:
21        printf("\nFriday");
22        break;
23     case 6:
24        printf("\nSaturday");
25        break;
26     case 7:
27        printf("\nSunday");
28        break;
29     default:
30        printf("\nInvalid");
31        break;
32    }
33 }
```

The output window shows the following text:

```
Enter: 4
Thursday

Process exited after 2.845 seconds with return value 0
Press any key to continue . . .
```

The IDE status bar shows 'Line: 13 Col: 19 Set: 0 Lines: 33 Length: 487 Insert Done parsing in 0 seconds'.

HOME SUBMISSION:

TASK:4

The screenshot shows a C++ program in Dev-C++ titled "Untitled1.cpp". The code is as follows:

```
1 #include <stdio.h>
2 int main()
3 {
4     int status;
5     int miles_flown;
6     int ticket_price;
7     printf("Are You a Frequent Flyer(0 for NO, 1 for yes): ");
8     scanf("%d", &status);
9     printf("How many mile You have flown: ");
10    scanf("%d", &miles_flown);
11    printf("Ticket Price: ");
12    scanf("%d", &ticket_price);
13    if (status == 1 && miles_flown > 50000 && ticket_price > 80000){
14        printf("\nYou have been upgraded to business Class!");
15    }
16    else {
17        printf("\nEconomy Class");
18    }
19 }
20
```

Two terminal windows are shown. The first terminal shows the output for a frequent flyer with 90000 miles flown and a ticket price of 90000, resulting in a business class upgrade. The second terminal shows the output for a frequent flyer with 120000 miles flown and a ticket price of 120000, resulting in an economy class ticket.

Compiler Output:

```
Compilation results...
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\Student26\Desktop\Untitled1.exe
- Output Size: 120,601,562 Bytes
- Compilation Time: 0.09s
```

TASK:5

The screenshot shows a C++ program in Dev-C++ titled "Untitled1.cpp". The code is as follows:

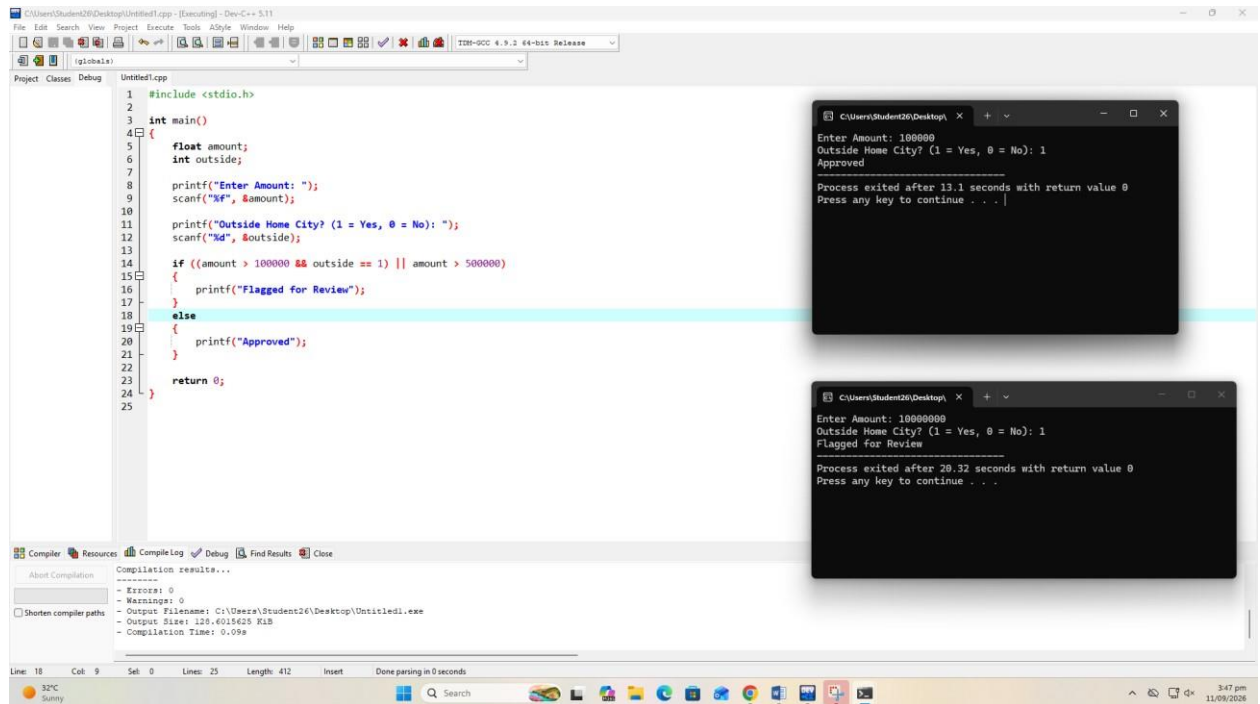
```
1 #include <stdio.h>
2 int main()
3 {
4     float hours_worked;
5     float rating;
6     printf("Hours Worked: ");
7     scanf("%f", &hours_worked);
8     printf("Rating: ");
9     scanf("%f", &rating);
10    if (hours_worked > 8 && rating >= 3 && rating <= 5) {
11        float new_hours = hours_worked - 8;
12        float cal1 = (500 * 1.5) * new_hours;
13        float cal2 = (500 * hours_worked);
14        float wage = cal1 + cal2;
15        printf("\nYour total daily wage (with overtime): %.2f", wage);
16    }
17    else {
18        float wage = 500 * hours_worked;
19        printf("\nYour Total Daily wage: %.2f", wage);
20    }
21 }
22
23
```

A terminal window is shown with the output for a worker with 9 hours worked and a rating of 4, resulting in a total daily wage of 5250.00.

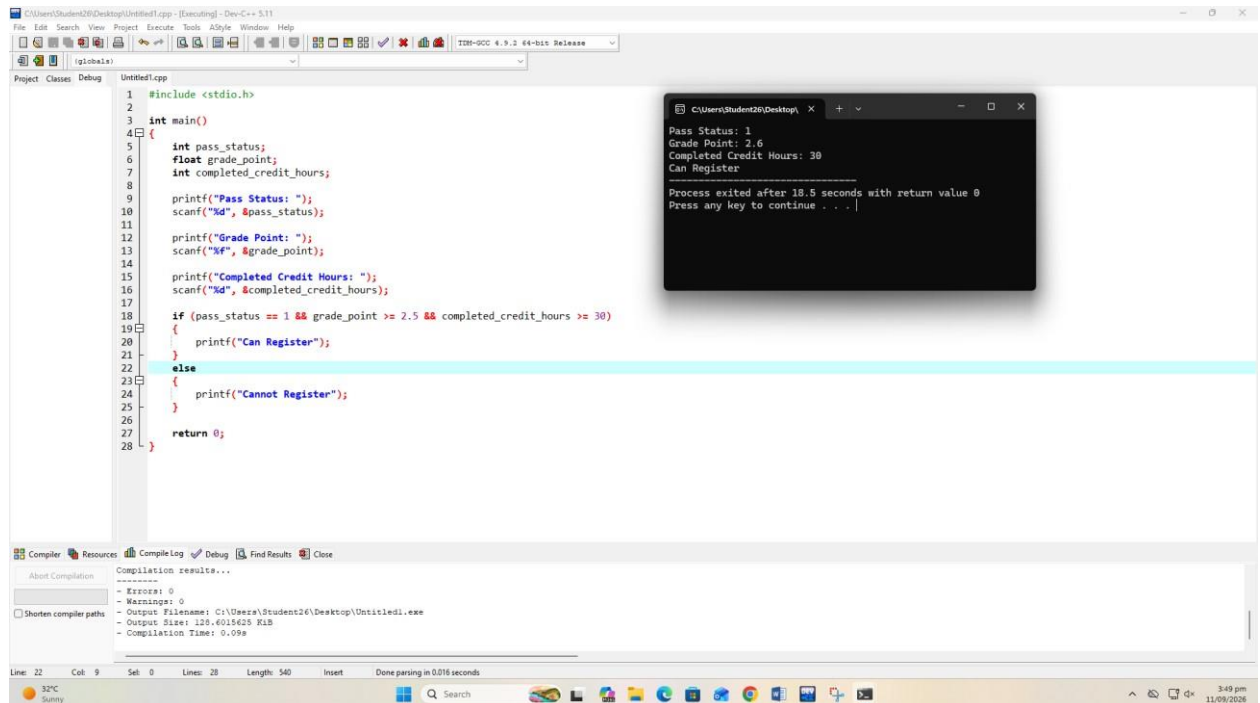
Compiler Output:

```
Compilation results...
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\Student26\Desktop\Untitled1.exe
- Output Size: 120,601,562 Bytes
- Compilation Time: 0.09s
```

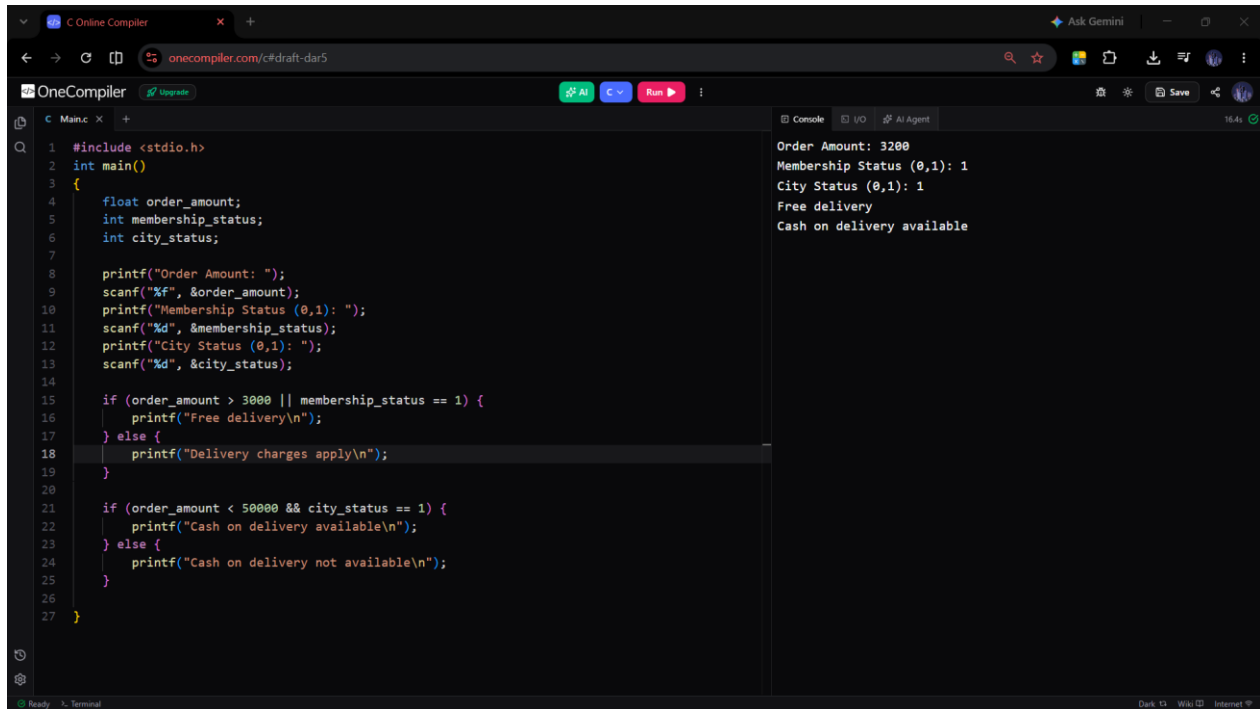
TASK:6



TASK:7



TASK:8



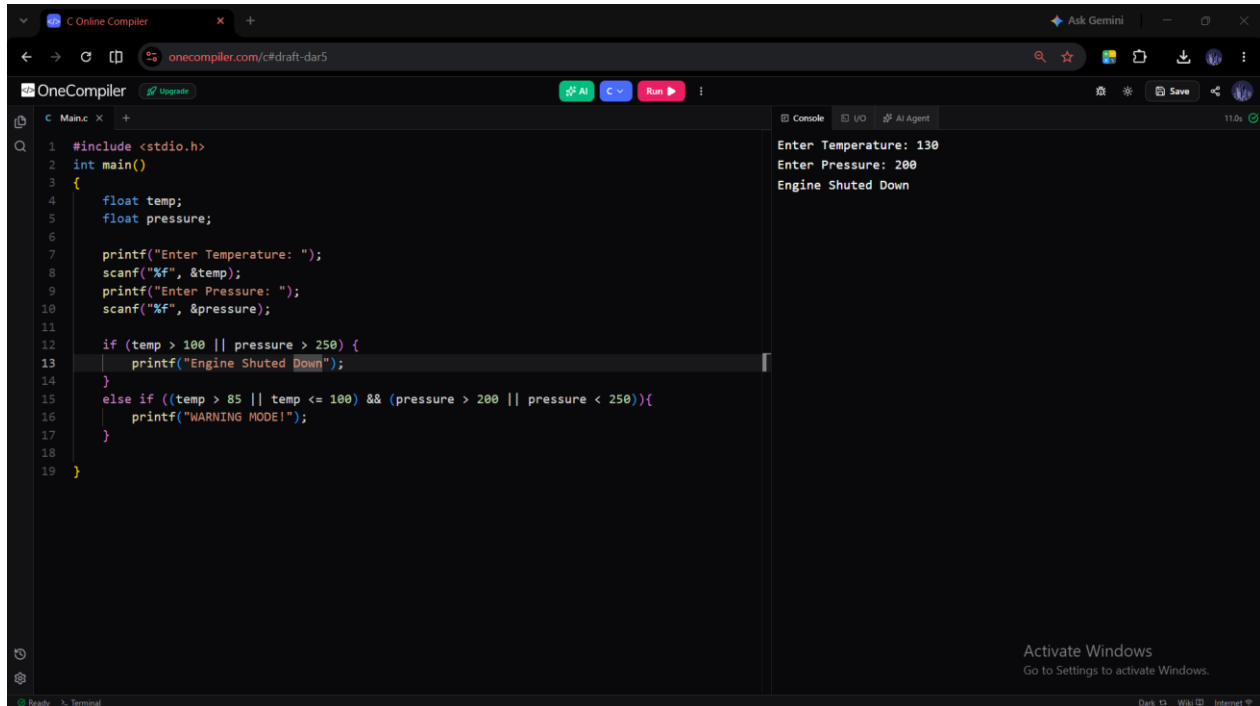
The screenshot shows the OneCompiler web interface. The code editor on the left contains a C program that calculates delivery status based on order amount and membership status. The console on the right shows the output of the program.

```
1 #include <stdio.h>
2 int main()
3 {
4     float order_amount;
5     int membership_status;
6     int city_status;
7
8     printf("Order Amount: ");
9     scanf("%f", &order_amount);
10    printf("Membership Status (0,1): ");
11    scanf("%d", &membership_status);
12    printf("City Status (0,1): ");
13    scanf("%d", &city_status);
14
15    if (order_amount > 3000 || membership_status == 1) {
16        printf("Free delivery\n");
17    } else {
18        printf("Delivery charges apply\n");
19    }
20
21    if (order_amount < 50000 && city_status == 1) {
22        printf("Cash on delivery available\n");
23    } else {
24        printf("Cash on delivery not available\n");
25    }
26
27 }
```

Console Output:

```
Order Amount: 3200
Membership Status (0,1): 1
City Status (0,1): 1
Free delivery
Cash on delivery available
```

TASK:9



The screenshot shows the OneCompiler web interface. The code editor on the left contains a C program that checks engine status based on temperature and pressure. The console on the right shows the output of the program.

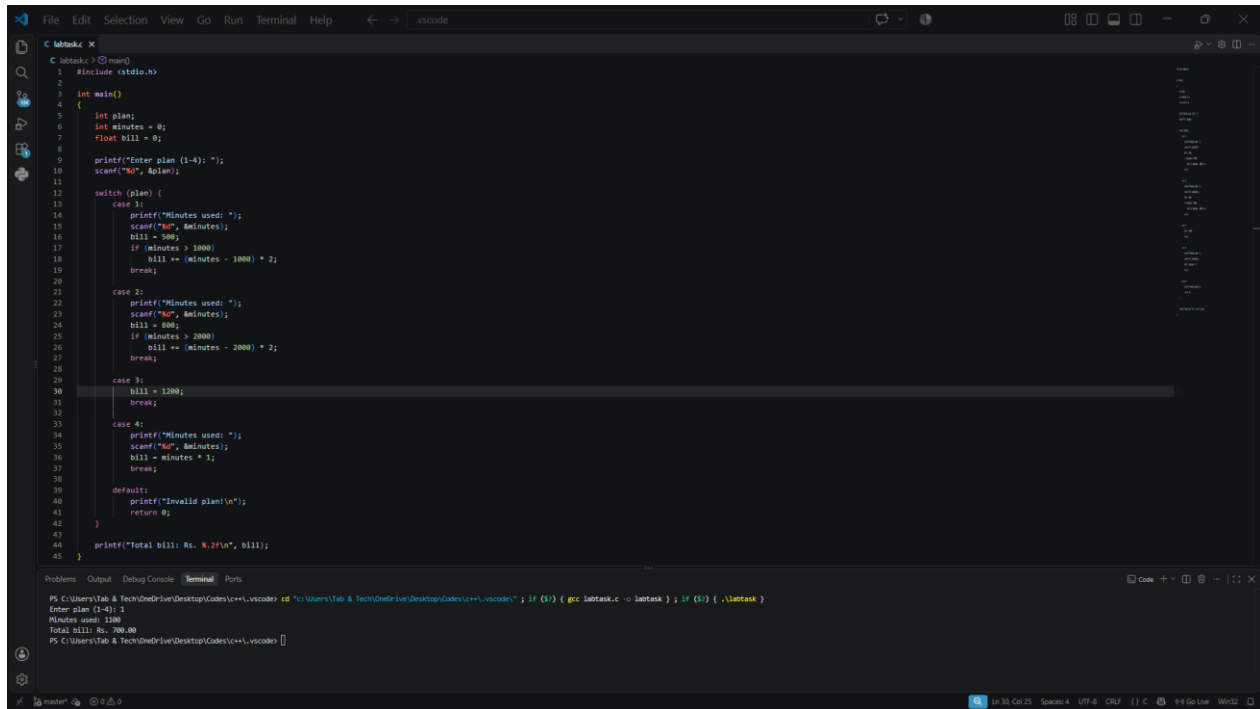
```
1 #include <stdio.h>
2 int main()
3 {
4     float temp;
5     float pressure;
6
7     printf("Enter Temperature: ");
8     scanf("%f", &temp);
9     printf("Enter Pressure: ");
10    scanf("%f", &pressure);
11
12    if (temp > 100 || pressure > 250) {
13        printf("Engine Shuted Down");
14    }
15    else if ((temp > 85 || temp <= 100) && (pressure > 200 || pressure < 250)){
16        printf("WARNING MODE!");
17    }
18
19 }
```

Console Output:

```
Enter Temperature: 130
Enter Pressure: 200
Engine Shuted Down
```

Activate Windows
Go to Settings to activate Windows.

TASK:10

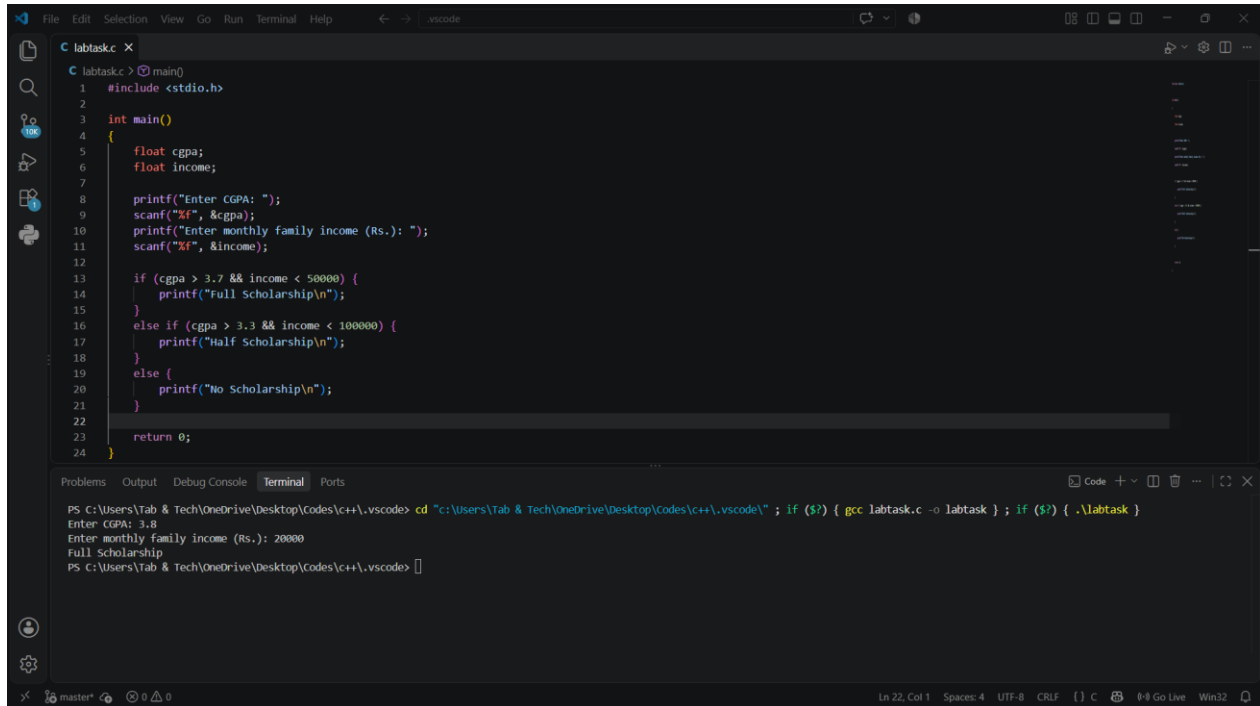


```
1 #include <stdio.h>
2
3 int main()
4 {
5     int plan;
6     int minutes = 0;
7     float bill = 0;
8
9     printf("Enter plan (1-4): ");
10    scanf("%d", &plan);
11
12    switch (plan) {
13        case 1:
14            printf("Minutes used: ");
15            scanf("%d", &minutes);
16            bill = 500;
17            if (minutes > 1000)
18                bill += (minutes - 1000) * 2;
19            break;
20
21        case 2:
22            printf("Minutes used: ");
23            scanf("%d", &minutes);
24            bill = 800;
25            if (minutes > 2000)
26                bill += (minutes - 2000) * 2;
27            break;
28
29        case 3:
30            bill = 1200;
31            break;
32
33        case 4:
34            printf("Minutes used: ");
35            scanf("%d", &minutes);
36            bill = minutes * 1;
37            break;
38
39        default:
40            printf("Invalid plan\n");
41            return 0;
42    }
43
44    printf("Total bill: Rs. %.2f\n", bill);
45 }
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Tab & Tech\OneDrive\Desktop\Codes\c++\.vscode> cd "c:\Users\Tab & Tech\OneDrive\Desktop\Codes\c++\.vscode\"; if ($?) { gcc labtask.c -o labtask }; if ($?) { .\labtask }
Enter plan (1-4): 1
Minutes used: 1200
Total bill: Rs. 700.00
PS C:\Users\Tab & Tech\OneDrive\Desktop\Codes\c++\.vscode>
```

TASK:11

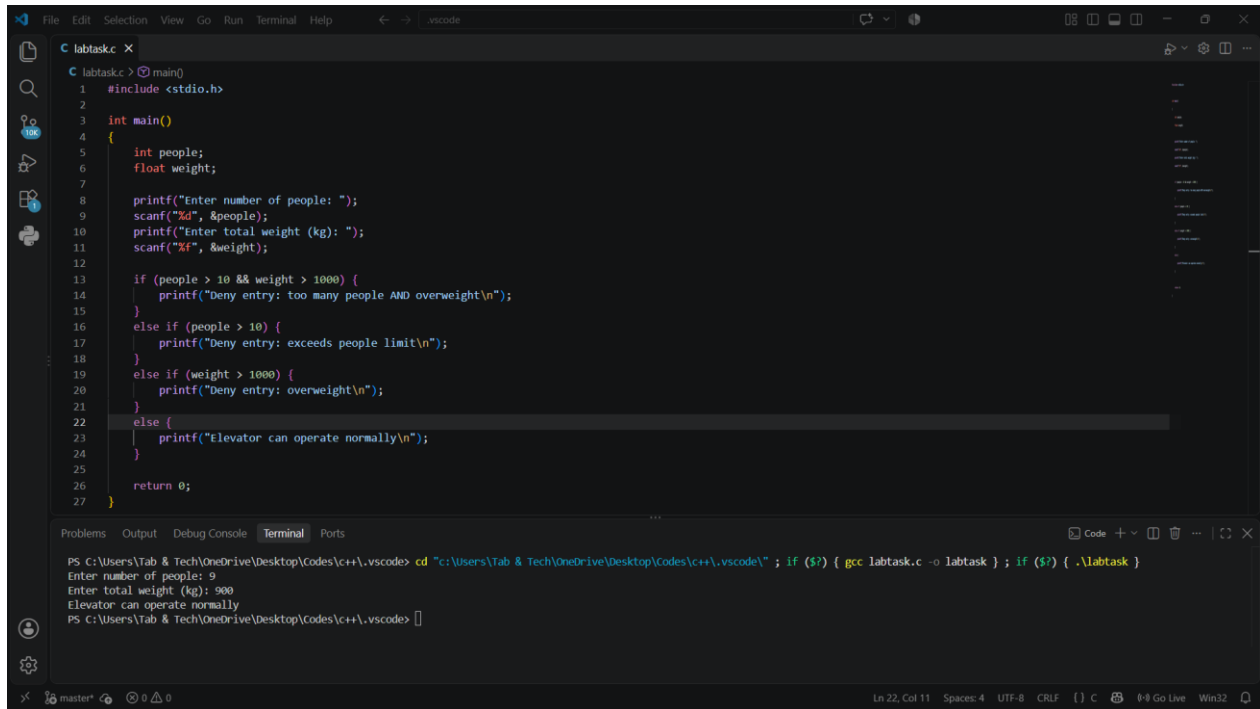


```
1 #include <stdio.h>
2
3 int main()
4 {
5     float cgpa;
6     float income;
7
8     printf("Enter CGPA: ");
9     scanf("%f", &cgpa);
10    printf("Enter monthly family income (Rs.): ");
11    scanf("%f", &income);
12
13    if (cgpa > 3.7 && income < 50000) {
14        printf("Full Scholarship\n");
15    }
16    else if (cgpa > 3.3 && income < 100000) {
17        printf("Half Scholarship\n");
18    }
19    else {
20        printf("No Scholarship\n");
21    }
22
23    return 0;
24 }
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Tab & Tech\OneDrive\Desktop\Codes\c++\.vscode> cd "c:\Users\Tab & Tech\OneDrive\Desktop\Codes\c++\.vscode\"; if ($?) { gcc labtask.c -o labtask }; if ($?) { .\labtask }
Enter CGPA: 3.8
Enter monthly family income (Rs.): 20000
Full Scholarship
PS C:\Users\Tab & Tech\OneDrive\Desktop\Codes\c++\.vscode>
```

TASK:12



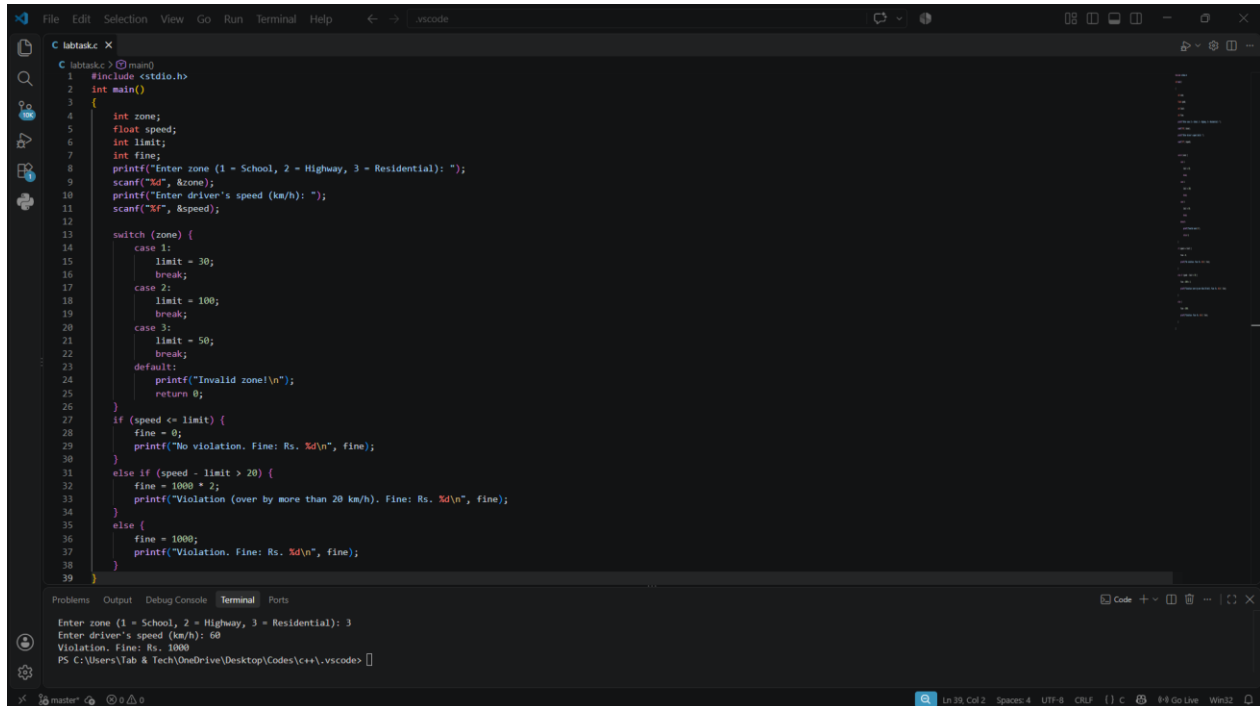
```
1 #include <stdio.h>
2
3 int main()
4 {
5     int people;
6     float weight;
7
8     printf("Enter number of people: ");
9     scanf("%d", &people);
10    printf("Enter total weight (kg): ");
11    scanf("%f", &weight);
12
13    if (people > 10 && weight > 1000) {
14        printf("Deny entry: too many people AND overweight\n");
15    }
16    else if (people > 10) {
17        printf("Deny entry: exceeds people limit\n");
18    }
19    else if (weight > 1000) {
20        printf("Deny entry: overweight\n");
21    }
22    else {
23        printf("Elevator can operate normally\n");
24    }
25
26    return 0;
27 }
```

Problems Output Debug Console Terminal Ports

PS C:\Users\Tab & Tech\OneDrive\Desktop\Codes\c++\.vscode> cd "c:\Users\Tab & Tech\OneDrive\Desktop\Codes\c++\.vscode\" ; if (\$?) { gcc labtask.c -o labtask } ; if (\$?) { .\labtask }

Enter number of people: 9
Enter total weight (kg): 900
Elevator can operate normally
PS C:\Users\Tab & Tech\OneDrive\Desktop\Codes\c++\.vscode>

TASK:13



```
1 #include <stdio.h>
2 int main()
3 {
4     int zone;
5     float speed;
6     int limit;
7     int fine;
8     printf("Enter zone (1 = School, 2 = Highway, 3 = Residential): ");
9     scanf("%d", &zone);
10    printf("Enter driver's speed (km/h): ");
11    scanf("%f", &speed);
12
13    switch (zone) {
14        case 1:
15            limit = 30;
16            break;
17        case 2:
18            limit = 100;
19            break;
20        case 3:
21            limit = 50;
22            break;
23        default:
24            printf("Invalid zone!\n");
25            return 0;
26    }
27
28    if (speed <= limit) {
29        fine = 0;
30        printf("No violation. Fine: Rs. %d\n", fine);
31    }
32    else if (speed - limit > 20) {
33        fine = 1000 * 2;
34        printf("Violation (over by more than 20 km/h). Fine: Rs. %d\n", fine);
35    }
36    else {
37        fine = 1000;
38        printf("Violation. Fine: Rs. %d\n", fine);
39    }
40 }
```

Problems Output Debug Console Terminal Ports

Enter zone (1 = School, 2 = Highway, 3 = Residential): 3
Enter driver's speed (km/h): 60
Violation. Fine: Rs. 1000
PS C:\Users\Tab & Tech\OneDrive\Desktop\Codes\c++\.vscode>

